

Internal Time as Local Memory: Explicit Termination in Anonymous Dynamic Binary Consensus

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Abstract

We study deterministic binary consensus in anonymous synchronous 1-interval-connected dynamic networks under one-bit broadcast-counting communication. The paper compares three ways of certifying that enough information-propagation time has elapsed. First, stabilization is size-oblivious: a two-state monotone flooding rule computes the global AND, so one bit of persistent local state is necessary and sufficient. Second, in a clockless model with time-homogeneous transitions, explicit termination has a logarithmic local-memory cost. If the network size n is known non-uniformly, then $M_{\text{term}}^{\text{size}}(n) = \Theta(\log n)$. If instead the size is unknown but a temporal dissemination bound (dynamic diameter) D is known non-uniformly, then $M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D + 1))$. Both lower bounds follow from the same mechanism: a split-cycle indistinguishability argument forces one homogeneous execution to wait for many rounds, and a time-homogeneous deterministic trajectory cannot repeat a pre-final state without pumping to an earlier halt. Matching upper bounds use ordinary one-bit flooding plus an internal round counter. Third, when the current global round number is supplied for free as an external input, the counter disappears and known- D explicit termination uses only $O(1)$ persistent states. Thus, within this narrow leaderless anonymous consensus model, the logarithmic termination memory is not needed to represent the consensus value; it represents elapsed-time evidence that must otherwise be supplied by a clock. We do not claim novelty for flooding, locality, dynamic-diameter notions, or finite-state pumping individually; the claim is restricted to the resulting local-space characterizations and their comparison.

Keywords. anonymous dynamic networks; binary consensus; explicit termination; stabilization; dynamic diameter; local memory; one-bit communication; finite-state computation; round clock.

1 Introduction

Anonymous dynamic networks combine severe symmetry with adversarially changing communication. Recent work shows that substantial deterministic computation remains possible even when every agent broadcasts only one bit per round and observes only the number of neighbors broadcasting each bit value [1]. A separate and well-established distinction is that between *stabilizing* computation, where outputs eventually become permanently correct, and *terminating* computation, where processes must also detect completion and stop [4, 3].

This paper asks a narrower space-complexity question. Suppose the value to be computed is only binary consensus, and suppose the value itself is easy to propagate. How much persistent local state is needed to certify that enough time has elapsed to halt?

The distinction matters because synchronization and knowledge are not the same resource. In the clockless model studied here, agents act in synchronous rounds but their transition rules do not receive the current round number as a free input. If an agent must distinguish round 3 from round 300, that distinction must be represented in its local state. We compare this with a free-clock variant in which the transition rule may inspect the global round number without storing it.

Our results form one theorem family. Stabilizing binary consensus needs exactly one bit and no global parameter. For explicit termination, knowing the network size n gives local-memory complexity $\Theta(\log n)$. More generally, if n is unknown but agents are given a common upper bound D on temporal dissemination time, the correct parameter becomes D and the complexity is $\Theta(\log(D + 1))$. Finally, if the same known- D algorithm is supplied with a free global round clock, only constant persistent memory is needed.

The proof mechanism is shared rather than duplicated. A static cycle with two large monochromatic input regions forces one of the two homogeneous executions to wait until information could have crossed a suitable radius. On a homogeneous cycle, all agents follow the same deterministic state trajectory. With no external time input, a state repetition before the first final state would repeat forever or reproduce the final state earlier, contradicting minimality of the halting time. Therefore waiting for T rounds requires $T + 1$ distinct local states.

Contributions.

1. **Size-oblivious stabilization.** Binary consensus stabilizes with exactly one bit of local state under one-bit broadcast-counting communication.
2. **A common waiting-to-space principle.** Split-cycle indistinguishability combined with homogeneous-state pumping converts a forced waiting time into a lower bound on persistent local states.
3. **Known-size termination.** For $n \geq 4$, any known- n clockless terminating algorithm needs at least $\lfloor (n+2)/4 \rfloor + 1$ states on some execution, while $2n$ states suffice. Hence $M_{\text{term}}^{\text{size}}(n) = \Theta(\log n)$.
4. **Known-diameter termination.** If n is unknown but a temporal dissemination bound D is compiled into the algorithm, then at least $\lceil D/2 \rceil + 1$ states are necessary and $2(D + 1)$ states suffice. Hence $M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D + 1))$.
5. **Free-clock separation.** With the same known D but with the global round number supplied for free, explicit termination uses $O(1)$ persistent states. Thus clocklessness incurs a $\Theta(\log(D + 1))$ local-memory tax in this model family.

We do not claim novelty for monotone flooding, one-hop locality, dynamic-diameter concepts, split-view indistinguishability, finite-state pumping, or the general distinction between stabilization and termination. The intended claim is only the combined local-space characterization for this narrow binary-consensus setting.

2 Model and definitions

2.1 Dynamic network

The system consists of a fixed set V of $|V| = n$ agents. Time proceeds in synchronous rounds $t = 1, 2, \dots$. In each round an adversary chooses a simple undirected graph

$$G_t = (V, E_t).$$

The dynamic network is *1-interval-connected* if every G_t is connected. Static connected graphs are therefore admissible special cases.

Agents are anonymous, have no identifiers, execute identical deterministic transition rules, and have no distinguished leader. Each agent v has an initial input $x_v \in \{0, 1\}$.

2.2 One-bit broadcast-counting communication

In each round every active agent broadcasts one bit to all current neighbors. An agent does not learn neighbor identities. Its communication observation is only the pair (c_0, c_1) , where c_b is the number of neighbors broadcasting bit b . This is the one-bit broadcast-counting model studied by Blanc, Di Luna, and Viglietta [1]. The pair (c_0, c_1) is a per-round observation; our space measure concerns persistent local state.

2.3 Clockless local state

For a fixed non-uniform parameter p (for example $p = n$ or $p = D$), let A_p have local state set Q_p . Its broadcast bit is

$$\sigma_p : Q_p \rightarrow \{0, 1\},$$

and its next-state function is

$$\delta_p : Q_p \times \mathbb{N} \times \mathbb{N} \rightarrow Q_p.$$

The output is $\text{out}_p : Q_p \rightarrow \{0, 1\}$.

Crucially, there is *no free external round clock*: neither σ_p nor δ_p receives t as an argument. Synchronous rounds determine when transitions occur, but the numerical value of the current round must be encoded in local state if the algorithm needs it. We call such rules *time-homogeneous* or *clockless*.

The local-space cost is

$$M(A_p) = \lceil \log_2 |Q_p| \rceil,$$

with the output information included in Q_p . The description size of the transition table is not charged. Thus a known parameter may be compiled non-uniformly into the algorithm family without itself occupying working memory. Parameter knowledge and elapsed-time knowledge are therefore distinct: the transition table may depend on D , while the current round still cannot be read unless it is represented in state.

2.4 Free-clock variant

For comparison, in the *free-clock* model the transition rule may also inspect the current global round number:

$$\delta_p^{\text{clk}} : Q_p \times \mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow Q_p.$$

Allowing σ_p to depend on t would not affect our upper bound. The round number is treated as an external input and is not charged to persistent local memory.

2.5 Stabilization and explicit termination

An execution solves *stabilizing binary consensus* if there is a finite T after which all outputs agree forever, and unanimity validity holds: if all inputs equal b , the stabilized output is b .

For explicit termination, let $F_p \subseteq Q_p$ be the set of final states. A final state is permanent: once entered, the local state and output never change. An execution solves *explicitly terminating*

binary consensus if every agent eventually enters a final state, all final outputs agree, and unanimity validity holds. Our lower bounds use only unanimity validity, so they also hold under stronger usual validity conditions.

2.6 Temporal dissemination bound

For a dynamic graph sequence $\mathcal{G} = (G_1, G_2, \dots)$, a start round s , and a node u , define causal reachability recursively by

$$\begin{aligned} R_0^s(u) &= \{u\}, \\ R_{i+1}^s(u) &= R_i^s(u) \cup N_{G_{s+i+1}}(R_i^s(u)), \end{aligned}$$

where $N_G(S)$ denotes all vertices adjacent in G to some vertex of S . We say that $\mathcal{G}$ has *dynamic diameter at most D* if

$$R_D^s(u) = V$$

for every node u and every start round s . Equivalently, information originating at any node can causally reach every node within D rounds, regardless of the start time. For a static connected graph, this is its ordinary graph diameter.

In the known- D regime, one algorithm A_D must work for every network size n and every admissible dynamic graph sequence of dynamic diameter at most D . The size n itself is not supplied.

3 Constant-state stabilization

Proposition 1. *Binary consensus can be stabilized with exactly one bit of persistent local state, without knowing n or D .*

Proof. Each agent keeps one bit $z_v \in \{0, 1\}$, initially $z_v(0) = x_v$. It broadcasts z_v . After each round it sets

$$z_v(t+1) = z_v(t) \wedge \bigwedge_{u \in N_v(t+1)} z_u(t).$$

Thus an agent changes from 1 to 0 if it already has 0 or hears at least one 0, and never changes back.

Let $S_t = \{v : z_v(t) = 0\}$. Then $S_t \subseteq S_{t+1}$. If $\emptyset \neq S_t \neq V$, connectivity of the next communication graph gives an edge crossing $(S_t, V \setminus S_t)$, so at least one new node joins S_{t+1} . Hence any initial 0 eventually reaches every node, while an all-1 input never creates a 0. The stabilized value is the global AND.

Two states suffice. Conversely, unanimous 0 and unanimous 1 executions must eventually expose different outputs, so at least two distinguishable states are necessary. Therefore

$$\boxed{M_{\text{stab}} = 1.}$$

□

Remark 1. The computation may take many rounds, but stabilization does not require any node to certify that the final propagation round has already occurred.

4 The common lower-bound mechanism

The lower bounds for known size and known dynamic diameter are two applications of the same argument.

4.1 Locality on a static cycle

Let C_m be a static cycle on $m \geq 3$ vertices.

Lemma 1 (Radius- t locality). *After t rounds on C_m , the state of a node v is determined entirely by the initial states within graph distance at most t from v .*

Proof. By induction on t . At $t = 0$ only the initial state of v matters. In round $t + 1$, the new state of v depends on its previous state and on messages from its two neighbors. By the induction hypothesis, those quantities depend only on initial states within distance at most $t + 1$ of v . $\square$

The lemma is ordinary one-hop causality and does not use the one-bit restriction.

4.2 Homogeneous executions

Fix a clockless deterministic algorithm A_p and a static cycle on which it is required to terminate. For $b \in \{0, 1\}$, let E_b be the homogeneous execution in which every input is b . By anonymity and symmetry, all nodes have the same local state at every time. Write the common trajectory as

$$q_0^{(b)}, q_1^{(b)}, q_2^{(b)}, \dots$$

and let

$$\tau_b = \min\{t : q_t^{(b)} \in F_p\}$$

be the first final-state time. Unanimity validity gives final outputs 0 and 1 in E_0 and E_1 , respectively.

Lemma 2 (Split-cycle waiting). *Suppose a mixed input on a static cycle contains a zero-input node u and a one-input node v such that every node at distance strictly less than r from u has input 0, and every node at distance strictly less than r from v has input 1. Then*

$$\max\{\tau_0, \tau_1\} \geq r.$$

Proof. If $\tau_0 < r$ and $\tau_1 < r$, radius- t locality makes the complete local history of u through time τ_0 identical to the history in the all-zero execution. Hence u halts with output 0. Likewise, v halts with output 1. Because final outputs do not change, Agreement is violated. $\square$

Lemma 3 (No pre-termination repetition). *For either homogeneous execution E_b , the states*

$$q_0^{(b)}, q_1^{(b)}, \dots, q_{\tau_b}^{(b)}$$

are pairwise distinct.

Proof. On a homogeneous static cycle, every node broadcasts the same bit and sees the same degree-two count observation. Therefore the common state evolves under a deterministic unary map f_p that does not depend on time:

$$q_{t+1}^{(b)} = f_p(q_t^{(b)}).$$

If $q_i^{(b)} = q_j^{(b)}$ for $0 \leq i < j < \tau_b$, determinism gives $q_{i+k}^{(b)} = q_{j+k}^{(b)}$ for all subsequent k . Taking $k = \tau_b - j$ reproduces the first final state at time

$$i + \tau_b - j = \tau_b - (j - i) < \tau_b,$$

a contradiction. The first final state cannot equal an earlier state because finality is a property of the state itself. $\square$

Theorem 1 (Waiting requires states). *Under the hypotheses of the split-cycle waiting lemma, every clockless explicitly terminating algorithm has*

$$|Q_p| \geq r + 1.$$

Proof. One of the homogeneous executions has $\tau_b \geq r$, and its states through the first final state are pairwise distinct. $\square$

Remark 2. This theorem is the common engine for the n -parameterized and D -parameterized results below. The memory lower bound comes from representing distinguishable stages of waiting, not from the information content of the binary consensus value.

5 Known network size: $\Theta(\log n)$ memory

Let A_n denote a clockless algorithm family whose transition table may depend on the common known size n .

5.1 Lower bound

Theorem 2. *For every $n \geq 4$, any explicitly terminating binary-consensus algorithm A_n satisfies*

$$|Q_n| \geq \left\lfloor \frac{n+2}{4} \right\rfloor + 1.$$

Consequently,

$$M_{\text{term}}^{\text{size}}(n) = \Omega(\log n).$$

Proof. Use the admissible static cycle C_n . Assign one contiguous block of

$$L = \left\lfloor \frac{n}{2} \right\rfloor$$

vertices input 0 and the remaining vertices input 1. In a contiguous block of length L , the largest possible distance of a block vertex from the complement is

$$r_n = \left\lceil \frac{L}{2} \right\rceil = \left\lceil \frac{\lfloor n/2 \rfloor}{2} \right\rceil = \left\lfloor \frac{n+2}{4} \right\rfloor.$$

The larger block has at least the same depth. Hence we can choose opposite-input vertices u and v whose radius- $(r_n - 1)$ neighborhoods are monochromatic. The waiting-requires-states theorem gives $|Q_n| \geq r_n + 1$. $\square$

5.2 Upper bound

Proposition 2. *If n is compiled into A_n , explicit binary consensus is solvable with one-bit communication and at most $2n$ local states.*

Proof. Let $H = n - 1$. Each node stores a candidate bit $z \in \{0, 1\}$ and a phase $c \in \{0, 1, \dots, H\}$. State $c = H$ is final. Initially $z = x_v$ and $c = 0$. During each nonfinal round, nodes broadcast z , update z by monotone AND flooding, and increment c . For every pair state (z, c) , the output map is $\text{out}_n(z, c) = z$; when $c = H$ is reached, the corresponding state is final and retains the same output bit z . After the H th propagation round, the updated state has $c = H$ and is final.

Because every round graph is connected, if any node initially holds 0, the zero-set grows by at least one node per round until all n nodes hold 0. Thus $H = n - 1$ rounds suffice. There are $2(H + 1) = 2n$ states. $\square$

Theorem 3 (Known-size termination tax). *In the known-size clockless regime,*

$$M_{\text{term}}^{\text{size}}(n) = \Theta(\log n).$$

Together with size-oblivious stabilization,

$$M_{\text{stab}} = 1, \quad M_{\text{term}}^{\text{size}}(n) = \Theta(\log n).$$

6 Known dynamic diameter, unknown size: $\Theta(\log(D + 1))$ memory

We now remove knowledge of n . Instead, a common temporal dissemination bound D is compiled into an algorithm A_D , which must work for all sizes and all admissible sequences of dynamic diameter at most D .

6.1 Lower bound

Theorem 4. *For every $D \geq 1$, any clockless explicitly terminating binary-consensus algorithm A_D requires*

$$|Q_D| \geq \left\lceil \frac{D}{2} \right\rceil + 1.$$

Consequently,

$$M_{\text{term}}^{\text{diam}}(D) \geq \left\lceil \log_2 \left(\left\lceil \frac{D}{2} \right\rceil + 1 \right) \right\rceil.$$

Proof. For $D \geq 2$, consider the static cycle C_{2D} . Its ordinary graph diameter, and therefore its dynamic diameter under the definition above, is exactly D . Split the cycle into two contiguous blocks of equal size D , assigning input 0 to one block and input 1 to the other.

For a contiguous block of D vertices on this cycle, the maximum distance from a block vertex to the complement is exactly

$$r_D = \left\lceil \frac{D}{2} \right\rceil.$$

Indeed, if $D = 2k$, either central vertex is at distance k from the nearest opposite-input vertex; if $D = 2k + 1$, the unique central vertex is at distance $k + 1$. Thus both input blocks contain a vertex whose radius- $(r_D - 1)$ neighborhood is monochromatic. The waiting-requires-states theorem gives $|Q_D| \geq r_D + 1$.

For $D = 1$, the claimed bound is $|Q_1| \geq 2$, which already follows from unanimity validity: executions with unanimous inputs 0 and 1 must terminate with different output values, so their final states cannot be the same state. $\square$

6.2 Upper bound

Proposition 3. *If a dynamic-diameter bound D is compiled into A_D , explicit binary consensus is solvable without knowing n , using one-bit communication and at most $2(D + 1)$ local states.*

Proof. Each node stores a candidate bit $z \in \{0, 1\}$ and a phase $c \in \{0, 1, \dots, D\}$. State $c = D$ is final. Initially $z = x_v$ and $c = 0$. During each nonfinal round, the node broadcasts z , replaces z by the AND of its own candidate and the received candidates, and increments c . For every pair state (z, c) , the output map is $\text{out}_D(z, c) = z$; when $c = D$ is reached, the corresponding state is final and retains the same output bit z . After exactly D propagation rounds it enters the final state.

If all inputs are 1, every candidate remains 1. If some node starts with 0, monotone flooding propagates the existence of that 0 along all causal paths. By the definition of dynamic diameter, after D rounds every node has been causally reached by the original zero and therefore holds candidate 0. Agreement, unanimity validity, and termination follow. There are $2(D+1)$ states. $\square$

Theorem 5 (Known-diameter termination tax). *In the known- D , unknown- n , clockless regime,*

$$M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D+1)).$$

More explicitly,

$$\left\lceil \log_2 \left(\left\lceil \frac{D}{2} \right\rceil + 1 \right) \right\rceil \leq M_{\text{term}}^{\text{diam}}(D) \leq \lceil \log_2(2(D+1)) \rceil.$$

Remark 3. The theorem determines the asymptotic local-memory complexity. The remaining constant-factor gap in the number of states is not closed here.

7 A free global clock collapses the D -dependent memory

The previous lower bound uses time-homogeneity essentially. We now allow the current global round t to be read for free.

Proposition 4. *In the free-clock model with known D and unknown n , explicitly terminating binary consensus is solvable with $O(1)$ persistent local states and one-bit communication.*

Proof. Use a constant set of states encoding a live candidate bit and a final output bit, for example

$$\{\text{live0}, \text{live1}, \text{final0}, \text{final1}\}.$$

The output map returns the encoded candidate bit in live states and the encoded final bit in final states. For the first D rounds, the live candidate follows monotone AND flooding. At the end of round D , the transition rule, which can inspect the external round number, moves to the corresponding final state with the same output bit. Dynamic diameter at most D guarantees that every initial zero has reached every node by that time; if no zero exists, all candidates remain one. $\square$

Because unanimous inputs require distinguishable final outputs, at least a constant number of states is necessary. Therefore:

Theorem 6 (Free-clock regime). *For known D and unknown n ,*

$$M_{\text{term}}^{\text{diam,clk}}(D) = \Theta(1).$$

Corollary 1 (Clocklessness tax). *Within the known- D model family considered here,*

$$M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D+1)), \quad M_{\text{term}}^{\text{diam,clk}}(D) = \Theta(1).$$

Thus removing the free external round clock incurs a logarithmic persistent-memory tax.

Remark 4. This is not a claim that every distributed task without a free clock requires $\Omega(\log D)$ memory. The statement is specific to the present anonymous, leaderless, time-homogeneous binary-consensus model. Other resources or assumptions can change the tradeoff.

8 Why knowing a parameter matters

If neither a size bound nor a dissemination-time bound is supplied, the clockless uniform model has the usual indistinguishability obstruction.

Proposition 5. *No single deterministic size-oblivious and diameter-oblivious clockless algorithm can solve explicitly terminating binary consensus on all anonymous static cycles, even if its local state space is not a priori bounded by a function of the cycle size.*

Proof. In a homogeneous all-zero static cycle, all nodes follow the same local trajectory. If the algorithm terminates on every cycle, that uniform trajectory reaches a final zero state after some finite time τ_0 independent of the cycle size. Likewise the all-one homogeneous trajectory terminates after a finite size-independent time τ_1 .

Choose a cycle large enough to contain a zero block and a one block whose central vertices are farther than $\max\{\tau_0, \tau_1\}$ from the opposite input. Locality then makes those two vertices halt as in the two homogeneous executions, producing opposite final outputs. $\square$

The role of known n or known D is therefore not that either value must literally be stored in binary. Under our non-uniform convention the parameter can be compiled into the transition table. What still costs memory in the clockless model is distinguishing the successive elapsed-time stages needed before finalization.

9 Relation to prior work

Blanc, Di Luna, and Viglietta [1] initiate the systematic study of deterministic computation in anonymous dynamic networks with one-bit broadcast-counting communication. They show surprisingly broad computational power, including terminating computation under suitable size information and stabilizing computation without prior knowledge of n . The present paper does not improve their general computation machinery; it fixes the task to binary consensus and measures persistent local state across different sources of timing information.

Parzych and Daymude [2] prove that anonymous dynamic broadcast with termination detection requires $\Omega(\log n)$ memory per node in their setting. This establishes that logarithmic memory for termination is not unique to consensus. Broadcast and binary consensus are different tasks, however, and our lower bound is obtained directly from two opposite homogeneous executions and a split cycle rather than by reduction from broadcast.

Di Luna and Viglietta [3] study universal finite-state and self-stabilizing computation in anonymous dynamic networks without network-size knowledge, while Schmid, Felber, and Rincon Galeana [4] give a broad topological characterization of stabilizing consensus. These works reinforce that stabilization and explicit termination are genuinely different notions; no novelty is claimed here for that distinction itself.

Dynamic or causal network diameters are standard information-propagation parameters in the broader dynamic-consensus literature. For example, Biely, Robinson, and Schmid [5] analyze agreement in directed dynamic networks under connectivity and propagation assumptions, and later work uses dynamic causal diameter explicitly in solvability conditions. Our definition is the simpler undirected always-connected dissemination bound appropriate to monotone flooding.

Dolev and Hoch [6] give constant-space Byzantine consensus algorithms for a substantially different localized-adversary graph setting, with running time proportional to network diameter for graph families they consider. This is one reason we do not make the broad claim that “waiting for a diameter without a clock always costs $\log D$ memory.” The lower bound here depends on

leaderlessness, anonymity, homogeneous time-independent transitions, and the particular consensus requirement.

Kowalski and Mosteiro [7] study all-to-all communication in anonymous adversarial dynamic networks with logarithmic memory and communication. Their task reconstructs all inputs with multiplicities and is information-theoretically stronger than binary consensus; their logarithmic-space guarantees do not directly imply the narrow characterizations proved here.

A targeted literature screening did not reveal a direct statement combining the known- n , known- D /unknown- n , and free-clock local-memory characterizations for leaderless binary consensus in the one-bit broadcast-counting model. This is a screening statement, not a certification of novelty.

10 Claim boundary and limitations

The paper does *not* claim novelty for monotone flooding, one-hop-per-round locality, static-cycle indistinguishability, finite-state pumping, dynamic-diameter definitions, non-uniform compilation of known parameters, round counters, the general distinction between stabilization and termination, or previously known logarithmic-memory lower bounds for anonymous dynamic broadcast.

The claimed mathematical result is restricted to the following combined characterization for deterministic anonymous synchronous leaderless 1-interval-connected dynamic networks:

$$M_{\text{stab}} = 1,$$

$$M_{\text{term}}^{\text{size}}(n) = \Theta(\log n),$$

$$M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D + 1)),$$

and, when a global round number is supplied for free,

$$M_{\text{term}}^{\text{diam,clk}}(D) = \Theta(1).$$

The interpretation is likewise narrow: in this model, the extra persistent memory for explicit termination can be identified with internally representing elapsed-time evidence that a free clock would otherwise provide.

The remaining constant-factor gaps in state counts are not closed. Randomization, identifiers, leaders, Byzantine faults, asynchronous communication, weaker connectivity assumptions, alternative message models, and models in which halted processes disappear from communication are outside the present theorem statements. The lower-bound mechanism itself does not use the one-bit restriction, but the matching upper bounds are stated in the one-bit broadcast-counting model.

11 Conclusion

Binary consensus provides a small setting in which value propagation and termination certification can be separated cleanly. A single persistent bit suffices to make the consensus value stabilize. Explicitly knowing when to stop is more expensive when elapsed time is not externally available.

The same split-cycle and pumping mechanism yields logarithmic memory in two natural parameter regimes. Known network size gives

$$M_{\text{term}}^{\text{size}}(n) = \Theta(\log n),$$

while known temporal dissemination bound and unknown size give

$$M_{\text{term}}^{\text{diam}}(D) = \Theta(\log(D + 1)).$$

Supplying the global round number for free collapses the latter to constant persistent memory:

$$M_{\text{term}}^{\text{diam,clk}}(D) = \Theta(1).$$

The unifying point is therefore not merely that termination is harder than stabilization. In this clockless anonymous setting, a node that must halt needs a persistent representation of how far the execution has progressed. The logarithmic local-memory tax is the cost of that internal time certificate.

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